INCH-POUND
MIL-DTL-14997C (AR)
6 August 2009
SUPERSEDING
MIL-DTL-14997B (AR)
25 October 2007

DETAIL SPECIFICATION

PROJECTILE, 105MM, ILLUMINATING, M314A3 AND PROJECTILE, 105MM, INFRARED ILLUMINATING, M1064 LOADING ACCESSORIES

Reactivated after 06 August 2009 and may be used for new and existing designs and acquisitions

This specification is approved for use by the U.S. Army Armament Research, by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the loading accessories used in two types of 105MM illuminating projectiles.

1.2 Classification. The illuminating projectiles are classified as follows:

1.2.1 Projectile, 105MM, Illuminating, M314A3. The M314A3 is a semi-fixed visible-light illuminating cartridge for use with 105MM howitzer systems. The projectile is hereafter referred to as the M314A3.

1.2.2 Projectile, 105MM, Infrared Illuminating, M1064. The M1064 is a semi-fixed infrared illuminating cartridge for use with 105MM howitzer systems. The projectile is hereafter referred to as the M1064.

Comments, suggestions, or questions on this document should be addressed to the Commander, U.S. Army ARDEC, ATTN: RDAR-QES-E, Picatinny Arsenal, NJ 07806-5000 or emailed to ardecstdzn@conus.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

AMSC N/A FSC 1315

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:20Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are needed to meet the requirements specified in section 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to assure the completeness of this list, document users are cautioned that they must meet all requirements documents cited in section 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards and handbooks. The following specifications, standards and handbooks below form a part of this document to the extent specified herein.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-W-12332	-	Welding, Resistance, Spot, Seam, And Projection, for fabricating Assemblies of Low Carbon Steel.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1916	-	DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch or from the Standardization Documents Order Desk, 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications below form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC) DRAWINGS

9206744	-	Projectile, 105MM, Illuminating, M314A3, Loading Assembly
9206745	-	Gasket, baffle
9206749	-	Illuminating Canister Loading Assembly (Epoxy Liner)
9206750	-	Canister Base Assembly
9206751	-	Support, parachute
9206752	-	Base
9206753	-	Pin, swivel
9206754	-	Plate, base
9206755	-	Canister body assembly
9206756	-	Body
9206757	-	Rotation brake
9206761	-	Disc assembly
9206764	-	Nut, swivel pin
9206766	-	Parachute assembly
9206771	-	Plate, shroud
9206777	-	Plate, swivel pin
9227195	-	Bag, parachute

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:20Z
Check the source to verify that this is the current version before use.

9278187	-	Pin groove
13009272	-	Alternative Illuminating Canister Loading Assembly (Fiberboard Liner)
13009273	-	Alternative, Fiberboard Liner Canister Assembly
13009274	-	Alternative, fiberboard liner
13009275	-	Alternative, Plate, Baffle
13020792	-	Plug, IR
13020793	-	Liner, IR fiberboard
13020794	-	Infrared Canister Assembly
13020795	-	Infrared Candle Loading Assembly
13020796	-	Projectile, 105MM, Infrared Illuminating, M1064

(Copies of these drawings may be requested online at Pica.Drawing.Request@conus.army.mil or from the U.S. Army ARDEC, ATTN: RDAR-EIS-T, Picatinny, NJ 07806-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASME INTERNATIONAL

ASME B 46.1	-	Surface Texture (Surface Roughness, Waviness,and Lay)

(Copies of this document are available online at http://www.asme.org or from ASME Codes and Standards, 3 Park Avenue, MS/20N2, New York, NY 10016).

ASTM INTERNATIONAL

ASTM B 117	-	Standard Practice for Operating Salt Spray (Fog) Testing
ASTM D 644	-	Standard Test Method for Moisture Content of Paper and Paperboard by Oven Drying
ASTM D 4444	-	Standard Test Methods for Use and Calibration of Hand-Held Moisture Meters
ASTM E 18	-	Standard Test for Methods Rockwell Hardness Test of Metallic Materials
ASTM E 1417	-	Standard Practice for Liquid Penetrant Testing

(Copies of these documents are available online at http://www.astm.org or from ASTM International, 100 Barr Harbor Drive, West Conshohocken, PA 19428.)

2.4 Order of precedence. Unless otherwise noted herein or in contract, in the event of a conflict between the text or this specification and the references cited herein, the text of this specification takes precedence. Nothing in this document however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:20Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Required inspection.

3.1.1 First article inspection. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.2 Conformance. A sample shall be subjected to conformance inspection in accordance with 4.3.

3.2 Materials, parts and assemblies. All materials, parts, and assemblies shall be in accordance with the requirements in the drawings listed in paragraphs 4.3.2.1 through 4.3.2.20.

3.3 Strength of weld. The weld securing the brake and body shall withstand a pull as outlined on MIL-W-12332, Peel Tests.

3.4 Parachute support profile of support test. The maximum gap between edges shall not exceed 0.108 inches.

3.5 Workmanship.

3.5.1 Foreign matter. No part or assembly shall contain dirt, grease, chips, oil, rust, corrosion or other foreign matter.

3.5.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth.

3.5.3 Burr, sharp edges and cracks. No part shall have a burr, sharp edge and/or cracks which might interfere with the assembly or function of the round or which might be injurious to personnel handling the item.

3.5.4 Cloth. All cloth shall be free of tears, rips and foreign matter. The stitches in the cloth shall be neat and continuous without breaks, tears or foreign matter.

3.6 Salt spray. Inert component parts or subassemblies shall be subjected to a 5 percent salt spray test in accordance with ASTM B 117.

3.7 Liquid penetrant test. After final assembly and crimping of part 9206750 (canister base assembly), part 9206752 (base) shall be subjected to Liquid penetrant inspection. Upon examination, the base shall be free of cracks or discontinuities.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:20Z
Check the source to verify that this is the current version before use.

4. VERIFICATION

TABLE I. Requirement / verification cross-reference matrix.

Methods of Verification 1 – Analysis 2 – Demonstration 3 – Examination 4 – Test						Classes of Verification A – First Article B – Conformance		
Section 3 Reqts	Description	Verification Method				Verification Class		Section 4 Verification
		1	2	3	4	A	B	
3.1.1	First article inspection			X	X	X		4.2
3.1.2	Conformance inspection			X	X		X	4.3
3.2	Gasket baffle			X		X	X	4.3.2.1
3.2	Nut swivel pin			X	X	X	X	4.3.2.2/4.4.3/ 4.4.4
3.2	Plate swivel pin			X		X	X	4.3.2.3/4.4.3
3.2	Pin swivel			X		X	X	4.3.2.4/4.4.3
3.2	Plate base			X		X	X	4.3.2.5/4.4.3
3.2	Base			X		X	X	4.3.2.6/4.4.3
3.2	Canister base assembly			X	X	X	X	4.3.2.7/4.4.4
3.2	Body			X		X	X	4.3.2.8/4.4.1
3.2	Disc assembly			X		X	X	4.3.2.9
3.2	Rotation brake			X		X	X	4.3.2.10
3.2	Canister body assembly			X	X	X	X	4.3.2.11/ 4.4.2/4.4.4
3.2	Bag parachute			X		X	X	4.3.2.12
3.2	Parachute assembly			X		X	X	4.3.2.13
3.2	Support parachute			X		X	X	4.3.2.14/ 4.4.3/4.4.5
3.2	Plate shroud			X	X	X	X	4.3.2.15/ 4.4.3/4.4.4
3.2	Pin groove			X		X	X	4.3.2.16
3.2	Alternative, plate, baffle			X		X	X	4.3.2.17
3.2	Alternative fiberboard liner (M314A3)			X	X	X	X	4.3.2.18/4.4.6
3.2	Liner, IR fiberboard (M1064)			X	X	X	X	4.3.2.19/ 4.4.6
3.2	Plug, IR (M1064)			X		X	X	4.3.2.20
3.3	Hardness of canister body			X	X	X	X	4.4.1
3.3	Strength of weld			X	X	X	X	4.4.2
3.4	Parachute support profile			X	X	X	X	4.4.5
3.5.1	Foreign matter			X		X	X	4.3.2.1-4.3.2.20
3.5.2	Threads			X		X	X	4.3.2.2/4.3.2.4/ 4.3.2.7
3.5.3	Burr and sharp edges			X		X	X	4.3.2.2-4.3.2.8/ 4.3.2.10-4.3.2.11/ 4.3.2.13-4.3.2.17/ 4.3.2.20
3.5.4	Cloth			X		X	X	4.3.2.12/4.3.2.13
3.6	Salt spray				X	X	X	4.4.4
3.7	Liquid penetrant test				X	X	X	4.4.7

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:20Z
Check the source to verify that this is the current version before use.

4.1 Classification of inspection. The following types of inspection shall be conducted on this item:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. When specified, a sample shall be subjected to first article verification in accordance with Table II.

4.2.1 First article quantity. First article verification shall be performed on the quantity of items as indicated in table II.

4.2.2 First article inspections to be performed. The first article verification shall be performed in accordance with Table II.

4.2.3 First article rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:20Z
Check the source to verify that this is the current version before use.

Table II. First article inspection

EXAMINATION OR TEST	NUMBER OF SAMPLES	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Gasket, Baffle (Dwg. 92066745)			
Examination for defects	5	3.2	4.3.2.1
Nut, Swivel Pin (Dwg. 9206764)			
Examination for defects	5	3.2	4.3.2.2
Salt spray test	5	3.6	4.4.4
Plate Swivel Pin (Dwg. 9206777)			
Examination for defects	5	3.2	4.3.2.3
Pin. Swivel (Dwg. 9206753)			
Examination for defects	5	3.2	4.3.2.4
Plate, base (Dwg. 9206754)			
Examination for defects	5	3.2	4.3.2.5
Base (Dwg. 9206752)			
Examination for defects	5	3.2	4.3.2.6
Canister Base Assembly (Dwg. 9206750)			
Examination for defects	5	3.2	4.3.2.7
Salt spray	5	3.6	4.4.4
Liquid penetrant test	5	3.7	4.4.7

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:20Z
Check the source to verify that this is the current version before use.

Table II. First article inspection - Continued

EXAMINATION OR TEST	NUMBER OF SAMPLES	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Body (Dwg. 9206756)			
Examination for defects	5	3.2	4.3.2.8
Disc Assembly (Dwg. 9206761)			
Examination for defects	5	3.2	4.3.2.9
Rotation Brake (Dwg. 9206757)			
Examination for defects	5	3.2	4.3.2.10
Canister Body Assembly (Dwg. 9206755)			
Examination for defects	5	3.2	4.3.2.11
Salt spray	5	3.6	4.4.4
Test for adequate weld strength	5	3.3	4.4.2
Bag, Parachute (Dwg. 9227195)			
Examination for defects	5	3.2	4.3.2.12
Parachute Assembly (Dwg. 9206766)			
Examination for defects	5	3.2	4.3.2.13
Support, Parachute (Dwg. 9206751)			
Examination for defects	5	3.2	4.3.2.14
Profile of support test	5	3.4	4.4.5

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:20Z
Check the source to verify that this is the current version before use.

Table II. First article inspection - Continued

EXAMINATION OR TEST	NUMBER OF SAMPLES	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Plate, Shroud (Dwg. 9206771)			
Examination for defects	5	3.2	4.3.2.15
Salt spray	5	3.6	4.4.4
Pin, Groove (Dwg. 9279187)			
Examination for defects	5	3.2	4.3.2.16
Alternative, Plate Baffle (Dwg. 13009275)			
Examination for defects	10	3.2	4.3.2.17
Alternative Fiberboard Liner (M314A3) (Dwg. 13009274)			
Examination for defects	10	3.2	4.3.2.18
Fiberboard liner moisture content	5	3.2	4.4.6
Liner, IR fiberboard (M1064) (Dwg. 13020793)			
Examination for defects	10	3.2	4.3.2.19
Fiberboard liner moisture content	5	3.2	4.4.6
Plug, IR (M1064) (Dwg. 13020792)			
Examination for defects	10	3.2	4.3.2.20

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:20Z
Check the source to verify that this is the current version before use.

4.3 Conformance inspection.

4.3.1 Inspection lot formation. Lot numbering shall be in accordance with MIL-STD-1168. Inspection lots shall comply with the lot formation provisions of MIL-STD-1916.

4.3.2 Classification of characteristics.

a. Sampling requirements. Inspection sampling requirements for critical, major and minor characteristics are defined in MIL-STD-1916. Unless otherwise specified, inspection level IV shall be used for all characteristics defined as majors and inspection level II for all minor characteristics. Critical characteristics shall be addressed in accordance with MIL-STD-1916.

b. Conformance inspection. Conformation inspection shall be performed accordance with paragraph 4.3.2.1 through 4.3.2.20. For all conformance inspections the same sample specimen may be used for all non-destructive examinations or tests.

c. Alternative inspections. Alternative conformance inspections shall be submitted and approved in accordance with MIL-STD-1916.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:20Z
Check the source to verify that this is the current version before use.

Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.1	TITLE Gasket, baffle		SHEET 1 OF 1	DRAWING NUMBER 9206745
				NEXT HIGHER ASSEMBLY 9206744
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major	None defined			
Minor				
201	Presence of three (3) slots	Level II	3.2	Visual
202	Evidence of foreign matter	Level II	3.5.1	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:20Z
Check the source to verify that this is the current version before use.

Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.2	TITLE Nut, swivel pin		SHEET 1 OF 1	DRAWING NUMBER 9206764 NEXT HIGHER ASSEMBLY 9206746
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Pitch diameter of thread	Level IV	3.2	AIE
102	Minor diameter of thread	Level IV	3.2	AIE
103	Salt spray	4.4.4	3.6	4.4.4
Minor				
201	Distance across flats, max	Level II	3.2	AIE
202	Thickness	Level II	3.2	AIE
203	Surface finish improper	Level II	3.2	Visual/4.4.3
204	Protective finish inadequate	Level II	3.2	Visual
205	Evidence of damaged threads	Level II	3.5.2	Visual
206	Evidence of burr, sharp edges, and/or cracks	Level II	3.5.3	Visual
207	Evidence of foreign matter	Level II	3.5.1	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:20Z
Check the source to verify that this is the current version before use.

Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.3	TITLE Plate, swivel pin		SHEET 1 OF 1	DRAWING NUMBER 9206777
				NEXT HIGHER ASSEMBLY 9206746
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major	None defined			
Minor				
201	Inside diameter, min	Level II	3.2	AIE
202	Thickness, min	Level II	3.2	AIE
203	Protective finish inadequate	Level II	3.2	Visual
204	Surface finish improper	Level II	3.2	Visual/4.4.3
205	Evidence of foreign matter	Level II	3.5.1	Visual
206	Evidence of burr, sharp edges, and/or cracks	Level II	3.5.3	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:20Z
Check the source to verify that this is the current version before use.

Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.4	TITLE Pin, swivel		SHEET 1 OF 1	DRAWING NUMBER 9206753
				NEXT HIGHER ASSEMBLY 9206750
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Pitch diameter of thread	Level IV	3.2	AIE
102	Major diameter of thread	Level IV	3.2	AIE
Minor				
201	Large diameter	Level II	3.2	AIE
202	Diameter of ball (fwd and aft)	Level II	3.2	AIE
203	Total length	Level II	3.2	AIE
204	Distance from base to aft shoulder	Level II	3.2	AIE
205	Distance from base to ball race shoulder	Level II	3.2	AIE
206	Thickness of ball race flange	Level II	3.2	AIE
207	Full thread length, min	Level II	3.2	AIE
208	Surface finish improper	Level II	3.2	Visual/4.4.3
209	Evidence of foreign matter	Level II	3.5.1	Visual
210	Evidence of damaged threads	Level II	3.5.2	Visual
211	Evidence of burr, sharp edges, and/or cracks	Level II	3.5.3	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:20Z
Check the source to verify that this is the current version before use.

Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.5	TITLE Plate, base		SHEET 1 OF 1	DRAWING NUMBER 9206754
				NEXT HIGHER ASSEMBLY 9206750
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major	None defined			
Minor				
201	Outside diameter	Level II	3.2	AIE
202	Thickness	Level II	3.2	AIE
203	Protective finish inadequate	Level II	3.2	Visual
204	Surface finish improper	Level II	3.2	Visual/4.4.3
205	Evidence of foreign matter	Level II	3.5.1	Visual
206	Evidence of burr, sharp edges, and/or cracks	Level II	3.5.3	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:20Z
Check the source to verify that this is the current version before use.

Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.6	TITLE Base		SHEET 1 OF 1	DRAWING NUMBER 9206752
				NEXT HIGHER ASSEMBLY 9206750
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major	None defined			
Minor				
201	Outside diameter of flange	Level II	3.2	AIE
202	Outside diameter of base plate cavity	Level II	3.2	AIE
203	Inside diameter of base plate cavity	Level II	3.2	AIE
204	Inside diameter of bearing bore	Level II	3.2	AIE
205	Outside diameter of bearing bore boss	Level II	3.2	AIE
206	Diameter of hole	Level II	3.2	AIE
207	Total length	Level II	3.2	AIE
208	Distance from face to flange	Level II	3.2	AIE
209	Thickness of flange	Level Ii	3.2	AIE
210	Depth of base plate cavity	Level II	3.2	AIE
211	Depth of bearing bore	Level II	3.2	AIE
212	Radius missing	Level II	3.2	AIE
213	Protective finish inadequate	Level II	3.2	Visual
214	Surface finish improper	Level II	3.2	Visual/4.4.3
215	Evidence of foreign matter	Level II	3.5.1	Visual
216	Evidence of burr, sharp edges, and/or cracks	Level II	3.5.3	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:20Z
Check the source to verify that this is the current version before use.

Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.7	TITLE Canister base assembly		SHEET 1 OF 1	DRAWING NUMBER 9206750
				NEXT HIGHER ASSEMBLY 9206749/ 13020794/ 13009273
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Swivel does not rotate freely	Level IV	3.2	Manual
102	Any component missing or improperly assembled	Level IV	3.2	Visual
103	Crimp not 360 degrees	Level IV	3.2	Visual
104	Salt spray	4.4.4	3.6	4.4.4
105	Liquid penetrant test	4.4.7	3.7	4.4.7
Minor				
201	Evidence of foreign matter	Level II	3.5.1	Visual
202	Evidence of damaged threads	Level II	3.5.2	Visual
203	Evidence of burr, sharp edges, and/or cracks	Level II	3.5.3	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:20Z
Check the source to verify that this is the current version before use.

Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.8	TITLE Body		SHEET 1 OF 1	DRAWING NUMBER 9206756 NEXT HIGHER ASSEMBLY 9206755
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Hardness	4.4.1	3.2	4.4.1
Minor				
201	Inside diameter, min	Level II	3.2	AIE
202	Total length	Level II	3.2	AIE
203	Thickness, min	Level II	3.2	AIE
204	Diameter of hole, max.	Level II	3.2	AIE
205	Radius missing	Level II	3.2	Visual
206	Protective finish inadequate	Level II	3.2	Visual
207	Evidence of foreign matter	Level II	3.5.1	Visual
208	Evidence of burr, sharp edges, and/or cracks	Level II	3.5.3	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:20Z
Check the source to verify that this is the current version before use.

Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.9	TITLE Disc assembly		SHEET 1 OF 1	DRAWING NUMBER 9206761
				NEXT HIGHER ASSEMBLY 9206749/13009272/ 13020795
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major	None defined			
Minor				
201	Diameter, max.	Level II	3.2	AIE
202	Improperly assembled	Level II	3.2	Visual
203	Evidence of foreign matter	Level II	3.5.1	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:20Z
Check the source to verify that this is the current version before use.

Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.10	TITLE Rotation brake		SHEET 1 OF 1	DRAWING NUMBER 9206757
				NEXT HIGHER ASSEMBLY 9206755
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major	None defined			
Minor				
201	Total length	Level II	3.2	AIE
202	Thickness	Level II	3.2	AIE
203	Width	Level II	3.2	AIE
204	Profile of brake	Level II	3.2	AIE
205	Protective finish inadequate	Level II	3.2	Visual
206	Evidence of foreign matter	Level II	3.5.1	Visual
207	Evidence of burr, sharp edges, and/or cracks	Level II	3.5.3	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:20Z
Check the source to verify that this is the current version before use.

Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.11	TITLE Canister body assembly		SHEET 1 OF 1	DRAWING NUMBER 9206755
				NEXT HIGHER ASSEMBLY 9206749/13009273/ 13020794
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Any component missing or improper	Level IV	3.2	Visual/manual
102	Test for adequate weld strength	4.4.2	3.3	4.4.2
103	Salt spray	4.4.4	3.6	4.4.4
Minor				
201	Brake above tolerance	Level II	3.2	AIE
202	Protective finish inadequate (if applicable)	Level II	3.2	Visual
203	Evidence of foreign matter	Level II	3.5.1	Visual
204	Evidence of burr, sharp edges, and/or cracks	Level II	3.5.3	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:20Z
Check the source to verify that this is the current version before use.

Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.12	TITLE Bag, parachute		SHEET 1 OF 1	DRAWING NUMBER 9227195
				NEXT HIGHER ASSEMBLY 9206744/13020796
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major	None defined			
Minor				
201	Improper seams or stitching	Level II	3.2	Visual
202	Length from bottom of bag to start of closure stitching	Level II	3.2	AIE
203	Location of closure stitching from edge of top flap	Level II	3.2	AIE
204	Length of overlap at open end	Level II	3.2	AIE
205	Length of overlap at bottom (closed end)	Level II	3.2	AIE
206	Width at development	Level II	3.2	AIE
207	Length at development	Level II	3.2	AIE
208	Evidence of foreign matter	Level II	3.5.1	Visual
209	Evidence of damaged cloth	Level II	3.5.4	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:20Z
Check the source to verify that this is the current version before use.

Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.13	TITLE Parachute assembly		SHEET 1 OF 1	DRAWING NUMBER 9206766
				NEXT HIGHER ASSEMBLY 9206746
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major	None defined			
Minor				
201	Circumference of reinforcing band at hem, min	Level II	3.2	AIE
202	Length of parachute cord from hem to shroud wire	Level II	3.2	AIE
203	Variation in length of shroud wire	Level II	3.2	AIE
204	Any component missing or improperly assembled	Level II	3.2	Visual
205	Stitching improper, missing or broken	Level II	3.2	Visual
206	Hole or tear in canopy (greater than 1/8 in dia. Or length)	Level II	3.2	Visual
207	Shroud not sewn minimum distance onto radial tape	Level II	3.2	Scale
208	Crimp pull test	Level II	3.2	Tension Tester
209	Overlap of tape, minimum	Level II	3.2	AIE
210	Evidence of foreign matter	Level II	3.5.1	Visual
211	Evidence of burr, sharp edges, and/or cracks	Level II	3.5.3	Visual
212	Evidence of damaged cloth	Level II	3.5.4	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:20Z
Check the source to verify that this is the current version before use.

Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.14	TITLE Support, parachute		SHEET 1 OF 1	DRAWING NUMBER 9206751
				NEXT HIGHER ASSEMBLY 9206744/13020796
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major	None defined			
Minor				
201	Total length	Level II	3.2	AIE
202	Thickness	Level II	3.2	AIE
203	Profile of support test (radius and arc length)	4.4.5/3.4	3.2	AIE/4.4.5
204	Protective finish inadequate	Level II	3.2	Visual
205	Surface finish improper	Level II	3.2	Visual/4.4.3
206	Evidence of foreign matter	Level II	3.5.1	Visual
207	Evidence of burr, sharp edges, and/or cracks	Level II	3.5.3	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:20Z
Check the source to verify that this is the current version before use.

Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.15	TITLE Plate, shroud		SHEET 1 OF 1	DRAWING NUMBER 9206771
				NEXT HIGHER ASSEMBLY 9206766
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Salt spray	4.4.4	3.6	4.4.4
Minor				
201	Outside diameter	Level II	3.2	AIE
202	Inside diameter of center hole, min	Level II	3.2	AIE
203	Thickness	Level II	3.2	AIE
204	Diameter of shroud line holes, min	Level II	3.2	AIE
205	True portion of shroud line holes	Level II	3.2	AIE
206	Countersink missing	Level II	3.2	Visual
207	Surface finish improper	Level II	3.2	Visual/4.4.3
208	Protective finish inadequate	Level II	3.2	Visual
208	Evidence of foreign matter	Level II	3.5.1	Visual
209	Evidence of burr, sharp edges, and/or cracks	Level II	3.5.3	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:20Z
Check the source to verify that this is the current version before use.

Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.16	TITLE Pin, groove		SHEET 1 OF 1	DRAWING NUMBER 9278187
				NEXT HIGHER ASSEMBLY 9206744/13020796
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major	None defined			
Minor				
201	Outside diameter	Level II	3.2	AIE
202	Total length	Level II	3.2	AIE
203	Evidence of foreign matter	Level II	3.5.1	Visual
204	Evidence of burr, sharp edges, and/or cracks	Level II	3.5.3	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:20Z
Check the source to verify that this is the current version before use.

Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.17	TITLE Alternative, plate baffle		SHEET 1 OF 1	DRAWING NUMBER 13009275 NEXT HIGHER ASSEMBLY 9206744 / 13020796
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Diameter, max	Level IV	3.2	AIE
102	Holes number	Level IV	3.2	Visual
103	Holes size	Level IV	3.2	AIE
Minor				
201	Evidence of foreign matter	Level II	3.5.1	Visual
202	Evidence of burr, sharp edges, and/or cracks	Level II	3.5.3	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:20Z
Check the source to verify that this is the current version before use.

Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.18	TITLE Alternative, fiberboard liner (M314A3)		SHEET 1 OF 1	DRAWING NUMBER 13009274 NEXT HIGHER ASSEMBLY 13009272
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Outside diameter	Level IV	3.2	AIE
102	Overall length	Level IV	3.2	AIE
103	Thickness	Level IV	3.2	AIE
104	Moisture content	4.4.6.1	3.2	4.4.6
Minor				
201	Evidence of foreign matter	Level II	3.5.1	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:20Z
Check the source to verify that this is the current version before use.

Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.19	TITLE Liner, IR fiberboard (M1064)		SHEET 1 OF 1	DRAWING NUMBER 13020793 NEXT HIGHER ASSEMBLY 13020794
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Outside diameter	Level IV	3.2	AIE
102	Overall length	Level IV	3.2	AIE
103	Thickness	Level IV	3.2	AIE
104	Moisture content	4.4.6.1	3.2	4.4.6
Minor				
201	Evidence of foreign matter	Level II	3.5.1	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:20Z
Check the source to verify that this is the current version before use.

Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.20	TITLE Plug, IR (M1064)		SHEET 1 OF 1	DRAWING NUMBER 13020792 NEXT HIGHER ASSEMBLY 13020794
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Outside diameter	Level IV	3.2	AIE
102	Diameter at recess for base assembly	Level IV	3.2	AIE
103	Depth of recess for base assembly	Level IV	3.2	AIE
104	Length of step for fiberboard liner	Level IV	3.2	AIE
105	Diameter at step for fiberboard liner	Level IV	3.2	AIE
106	Distance to first stake notch	Level IV	3.2	AIE
107	Distance to second stake notch	Level IV	3.2	AIE
108	Distance to step for fiberboard liner	Level IV	3.2	AIE
Minor				
201	Evidence of foreign matter	Level II	3.5.1	Visual
202	Evidence of burr, sharp edges, and/or cracks	Level II	3.5.3	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:20Z
Check the source to verify that this is the current version before use.

4.4 Methods of inspection.

4.4.1 Hardness test of canister body (nondestructive test). A proper adapter shall be used so as to permit taking of Rockwell hardness readings on the outside surface of the canister body approximately one inch from the open end. Two hardness determinations, taken 180 degrees apart, shall be made on each sample canister body. Each determination shall consist of the average of two readings taken approximately 1/4 inch apart. Testing procedures shall be in accordance with ASTM E 18. Eight samples shall be randomly selected from each eight hours of production. If any of the hardness determinations taken on the sample canister bodies fail to comply with the requirements specified on drawing 9206756 and herein, the production represented by the sample shall be rejected.

4.4.2 Test for adequate weld strength. No more than one undersize weld button is permitted on each sample tested. More than one weld button which measures less than 0.13 inches diameter shall be cause for rejection of the lot. Three (3) assemblies shall be selected from each welding machine at the beginning of each day of production and every time the electrodes or machine adjustments are changed in order to qualify the machine. No machine shall be used for production unless all test specimens in the qualification sample comply with the requirement specified in 3.3.

4.4.3 Surface finish. The roughness comparison specimens prescribed in ASME B 46.1 shall be used as a basis for surface roughness determination. Four (4) power magnifications may be used in performing visual examination.

4.4.4 Salt spray test. Inert component parts or subassemblies selected for this test shall be subjected to a 5 percent salt spray test for 96 hours in accordance with ASTM B 117. After exposure, samples shall be examined for the effects of corrosion. Any evidence of corrosion on the test samples shall be a cause for rejection of the production represented by the sample. Two (2) samples shall be taken each week, one during the first 30 minutes of production and a second at any other time.

4.4.5 Profile of parachute support test. Testing shall be performed on four (4) samples simultaneously. Four (4) parts shall fit into a 3.090 +.001 inside diameter ring gage. With the four parts contacting at three (3) edges, the maximum gap between the remaining edges shall not exceed 0.108 inches. Failure to meet this requirement will be cause for rejection of al1 four (4) samples.

4.4.6 Fiberboard liner moisture content. The moisture content of the fiber board liner shall be determined in accordance with ASTM D 644 or ASTM D 4444. Failure of any sample to comply with the applicable requirement shall result in rejection of the inspection lot represented by the sample.

4.4.6.1 Fiberboard liner moisture content sampling. Five (5) fiberboard liners shall be randomly selected and tested for moisture content within 24 hours prior to use in production.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:20Z
Check the source to verify that this is the current version before use.

4.4.7 Liquid penetrant test. The item under test shall be inspected in accordance with ASTM E1417, Type 1, Method A, Sensitivity Level 3 (See 6.7.2). Testing and Monitoring (TAM) panels shall be used to check the effectiveness of the inspection process (See 6.7.1). Four (4) samples shall be randomly selected from each eight hours or one shift of production from the crimping machine. If two or more crimping machines are used in production, two (2) samples from each machine shall be randomly selected and evaluated. If any sample fails to comply with the requirements outlined in 3.7, the production represented by the sample shall be rejected. The tested samples shall be discarded or otherwise prevented from re-entering the product stream (See 6.7.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DOD personnel or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The projectile loading accessories contained in this specification are intended for use with Projectile, 105MM, Illuminating, M314A3 and Projectile, 105MM, Infrared Illuminating, M1064.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Requirements for submission of First Article sample (see 4.4).
c. Requirements for submittal of Acceptance Inspection Equipment (see 6.3).
d. Requirements for ammunition lot numbering.

6.3 Submission of contractor acceptance inspection equipment (AIE) designs for approval. Submit copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: RDAR-QEM-A, Picatinny, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Submission of ammunition data cards. In addition to the normal distribution of records for items procured by the Department of the Army, ammunition data cards should be forwarded

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:20Z
Check the source to verify that this is the current version before use.

to Commanding Officer, U.S Army ARDEC, ATTN: RDAR-MEM-C and RDAR-QEM-A, Picatinny Arsenal, NJ 07806-5000.

6.5 Automatic inspection equipment. All automatic test and inspection equipment when utilized by the contractor should be subjected to periodic verification or calibration checks during production. The procedures should include as a minimum: frequency, method of verification, and the method of retrieval of items produced between verification/calibration.

6.6 Automatic gaging. Facilities equipped with furnished automatic gaging capable of performing 100 percent inspection of any characteristic should utilize the capability in accordance with contract.

6.7 Liquid penetrant test.

6.7.1 Inspection standard. It is recommended that an approved standard manufactured from the item under test, or a known defective item be used as an additional verification of the inspection process.

6.7.2 Alternate procedure. Contractors may submit alternate methods of inspection for approval. In addition, procedures for masking and/or cleaning the tested base assemblies may also be submitted, demonstrating that the procedure used will not be detrimental to the part or mating components. This may allow for the reintroduction of tested samples into the product stream.

6.8 Definitions.

a. Acceptance Inspection Equipment (AIE). Any measuring device that is traceable to the national or international standard which is used to assure conformance of material to the contract requirements.

b. Verification methods.

Analysis. Use of analytical methods and mathematical formulas to determine if an item meets a requirement.

Demonstration. A functional test (for example ballistic firing) that provides verification that a requirement is met.

Examination. Inspections utilizing gaging or visual means to assure conformance to a requirement.

Test. The conduct of specific procedures and methods to provide data verifying a requirement is met.

6.9 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:20Z
Check the source to verify that this is the current version before use.

6.10 Subject term (keyword) listing.

Ammunition
Artillery
Munition

Custodian:
Army- AR

Preparing Activity:
Army-AR
Project (1315-2009-002)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:20Z
Check the source to verify that this is the current version before use.